Important Points

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Note:

Topologies → Physical topologies - mesh, star, bus etc
→ logical topologies - IEEE 802.3, IEEE 802.5

Physical topologies deals with high physically system is connected whereas logical topologies deals with how data is transmitted on LAN Network.

(i) Mesh topology

n devices → $\frac{n(n-1)}{2}$ lines

n-1 ports (to how many devices one device can send)

(ii) Star topology

n device → n droplines + 1 backbone cable (for ring)

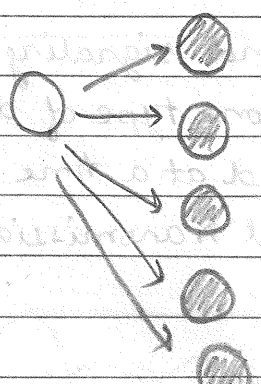
(iii) Star topology

n devices → n links - one port

Passive hub simply forwards data because it is non intelligent device.

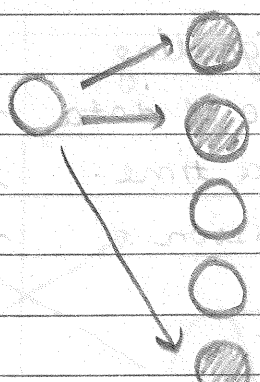
An active hub will make decision whether to block or forward data.

Broadcast device



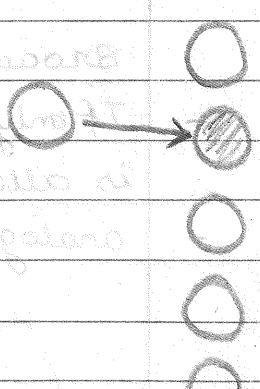
to all device on network

Multicast device



to only group on network

Unicast device



to only one device.

difference b/w circuit (CS) switching & packet (PS) switching.

(i) C.S is reliable, P.S is unreliable.

(ii) C.S is used for log messages, P.S is for short messages

(iii) C.S is slow, P.S is fast.

Collision Domain - the place or area where collision occur.

Hub is not a collision domain seperator that is if hub is used as center device, then each port has seperate collision Domain.

Broadcast Domain - The place or area where broadcast has been done.

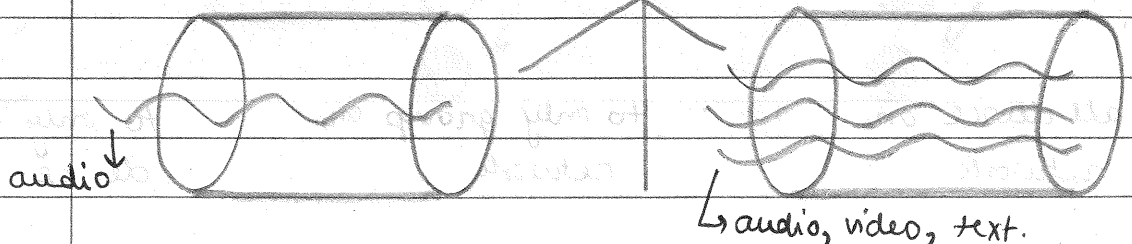
switch —access link→ computer

switch —translink→ switch.

10 base 2 → length is 200 m, bandwidth = 10 Mbps

10 base 5 → length is 500 m, bandwidth = 10 Mbps.

Broadband signaling	Baseband signaling.
- If more than one type of data is allowed at a time.	- if only one type of data is allowed at a time
- analog transmission	- digital transmission.



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Monitor is responsible for proper operation of ring. If no station is having data, token will simply circulate in ring. only those will pass data who will have token

Ring latency → time taken to take one rotation of ring

	IEEE 802.3 (ethernet)	IEEE 802.5 (ring)
(i)	It is non deterministic channel	(i) It is deterministic channel.
(ii)	There are no priorities	(ii) There are priorities
(iii)	connection protocal	(iii) connectionless protocal.
(iv)	It uses manchester encoding	(iv) It uses differential manchester encoding.

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LAN (Local area network)

- are private network, not subject to traffics or other regulatory controls.
- operate at relative high speed
- different types of Media Access control - Ethernet, token ring etc.
- it connects computer in single building or campus, ie they work in restricted geographical area.
- LAN are widely used to share resources like printer, shared hard-drive etc.
- connect all workstation in building to communicate with each other locally without internet access.

advantages: resource sharing, software application sharing, easy and cheap communication, centralised data, data security, Internet sharing.

disadvantage: high setup cost, privacy violations, data security threat, LAN maintenance, covers limited area.

MAN (metropolitan Area Network)

- generally covers town and cities (50 km).
- Communication medium used for MAN are optical fibre, cables etc.
- data rates are adequate for distributed communication system.

advantages: fast communication, good backbone for large network

disadvantage: more cable required, difficult to make system secure from hackers.

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WAN (wide area network).

- can be private or public leased network.
- communication medium used by WAN are PSTN or satellite links.
- generally covers large distances (states, country, continents)

advantage: everyone on network can use same data, covers large area, message can be sent very quickly on network, expensive things can be shared.

disadvantage: needs a good firewall to restrict outsiders, setting up network can be expensive, slow, security issue.

Wireless Network.

Morse code was used to implement wireless network.

System Interconnection	Wireless LAN	Wireless WAN.
- (short range communication) like using Bluetooth to connect various components like monitor, mouse etc - uses master-slave concept. system unit is master, mouse etc is slave.	- Installing Ethernet is considered trouble, hence standard for wireless LAN called IEEE 802.11 is widespread. - every computer has radio modem and antenna to communicate	- Radio network used for cellular telephone is an example of low-bandwidth wirelessWAN. - first generation → analog and voice. second generation → digital and voice. third generation → for both voice & data.

Services.: operations that a layer can provide to layer above in OSI model. It defines operation and states in a layer is ready to perform but doesnot specify anything about implementation of these operation.

Protocols: these are set of rules that govern messages or packets that are exchanged b/w server and client.

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Connection oriented & connectionless service.

(i) In connection oriented authentication is needed, while connectionless service doesnot need authentication

(ii) connection oriented checks message is recieved or not, if not sends it again whereas connectionless service doesnot guarantee a message delivery.

(iii) connection oriented is more reliable than connectionless.

(iv) connection oriented service interface is stream based and connectionless is message based.

(v) connection oriented example - TCP (Transmission control protocal), connectionless example - UDP (User Datagram Protocal).

Service Primitives.

A service is formally specified by set of primitives (operations) available to user process to access the service. These primitives tell the service to perform some action or report on an action taken by peer entity. If protocal stack is located in O.S. (operating system), the primitives are called system calls. These calls causes trap to kernel mode.

Types of service primitives

(i) Listen - block waiting for incoming message
(ii) connect - establish connection
(iii) Recieve - block waiting for incoming message
(iv) send - sending a message
(v) Disconnect - terminate.

} for connection oriented.

(i) UNIDATA - sends a packet of data
(ii) FACILITY, REPORT - enquiring about performance of network.

} for connectionless

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Network devices

(I) **Hub:** least expensive, least intelligent, least complicated
If a message comes in, it is sent to all other ports
Hub is a multiport repeater. Hub cannot filter data, so data packets are sent to all connected devices.

Types of Hub.

(a) Active Hub: have their own power supply (repeater + wiring center)
(b) Passive hub: collect power supply from active hub. (do not act as repeater)

Passive hub simply forwards data packets whereas active hub (multiport repeater) regenerates data bits in order to maintain strong signal.

(II) **Repeater:** operates at physical layer. Its job is to regenerate the signal over same network before signal becomes weak or corrupted.
They donot amplify the signal. (2 device port)

(III) **Bridge:** operates at data link layer. A bridge is a repeater, with add functionality of filtering content by reading MAC address of source and destination.

Used for connecting two LAN working on same protocol.

Bridges are faster than router (2 port device)

Types of Bridge

(I) Transparent Bridge: stations are completely unaware of other bridge.
(II) Source Routing Bridge: routing operation is performed by source.

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(iv) Switch : multiport bridge with a buffer.
It is a datalink layer device. Switch can perform error checking before forwarding data ie it forwards only good packets to its port. In other words, switch divides collision domain of hosts, but broadcast domain remains same.

(v) Router : A router is a device like a switch that routes data packets based on their IP addresses.
Router is mainly Network layer device.
Router connects normally LANs and WANs together and have dynamically updating routing table
Router divide broadcast domains of hosts connected through it. They use protocols such as ICMP to communicate with each other.
Different topology and same protocal.

(vi) Gateway : is a passage to connect two network together that may work upon different networking model. They are also called protocol converters.
They operate at network layer.
Different topology and different protocal.

(vii) Brouter : (bridging router) is a device with combines feature of both bridge and router. Works at either data link layer or network layer.

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→ OSI model (open system Interconnection)
ISO (International organization of Standardization) has developed a standard to communicate with each other - system which are compatible)

OSI is seven layer architecture :

Application layer } manage
Presentation layer
Session layer
Transport layer → TCP segment / UDP datagram (gateway) ← all layer
Network layer → packets three layer → (router)
Datalink layer → frames two layer → (bridge)
Physical layer → bits one layer → (repeater / hub)

only important function of these layers

(I) Application layer provides services like http, ftp, smtp, DNS etc It deals with syntax (pulses) and semantics (meanings) of data. It holds mainly application program.

(II) Presentation layer performs data compression, encryption, conversion etc. It takes care of data to be sent in such a way that receiver will understand data.

(III) Session layer manages and synchronize conversation b/w two different application. Streams of data are marked and resynchronized to avoid data loss.

(IV) Transport layer provides flow control, error control, segmentation, congestion, splitting. It breaks message into smaller units so they can be handled more efficiently by network layer.

(v) Network layer: routing, fragmentation, traffic shapping, network controller.

(vi) Data link layer: flow control, error control, access control, MAC protocols, acknowledgement / non-acknowledgement.

(vii) Physical layer: activates, maintain, deactivates the physical connection, data encoding, converts analog / digital bits into electrical signal or optical signal, responsible for unstructured raw data over network.

Network layer → node to node delivery using router.
Transport layer → process to process or end to end delivery.

Merits of OSI Model.

(i) distinguish well b/w services, interfaces and protocols
(ii) protocols are well hidden
(iii) protocols can be changed by new protocols.
(iv) supports connection oriented as well as connectionless service.

Demerits of OSI Model.

(i) Model was devised (designed) before invention of protocols.
(ii) Fitting of protocol is tedious task.
(iii) It is just used as reference model.

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seven layer of OSI model has been reduced to five layers of TCP/IP model by providing functionalities of presentation & session layer in application layer.

wirless uses UDP protocal.

TCP/IP protocal:
- network model used in current Internet architecture.
- Protocals describe movement of data b/w source & destination
- Protocals offer naming and addressing scheme.

Protocals in TCP/IP
(i) Application - TELNET, FTP, SMTP, DNS
(ii) Transport - TCP, UDP
(iii) Network - IP

Network in TCP/IP
(i) Physical + Data link layer - ARPANET, SATNET, Packet Radio, LAN.

Features of TCP/IP
(i) flexible architecture ie adding more machine to network was easy.
(ii) network was robust and connection remained intact until source and destination machine are functioning

Merits of TCP/IP
(i) scalable, operated independently
(ii) client-server architecture
(iii) supports no. of routing protocal.
(iv) can be used to establish connection.

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Demerits of TCP/IP.

(i) doesnot guarantee delivery of packets
(ii) model cannot be used in other application
(iii) replacing protocol is not easy.
(iv) not clearly seperated services, interfaces and protocols.

File Transfer Protocol.

FTP is a client-server protocol that relies on two communications channel b/w client and server, a command channel for controlling conversations and a data channel for transmitting file content.

Using FTP, a client can upload, download, delete, rename, move, copy file on server.

A user typically needs to logon to FTP server, although some server make some or all of their content available without login, known as anonymous FTP.

FTP session work in passive or active modes.

In active mode, after a client initiates a session via a command channel request, server initiates a data connection back to client and begins transferiy data.

In passive mode, server instead uses command channel to send client information it needs to open a data channel. Client initiates all connections. It works well across firewall and NAT (Network Addrers Translation) gateways.

Users can work with FTP via commandline as well as GUI.

FTP is an application layer protocol which moves files b/w local and remote file system.

To transfer a file, 2 TCP connections are used by FTP in parallel →

(i) Control connection: for sending control information like user identification, password, commands. (Port no 21)

(ii) Data connection: for sending actual file. (Port no 20.)

FTP allows three types of data structure.

(i) File structure - no internal structure, file is continuous sequence of bytes.

(ii) Record structure - file is made up of sequential records.

(iii) Page structure - file is made up of independent indexed pages.

PPP (Point to Point protocol) (RFC 1661)
is a data link layer communication protocol used to establish a direct connection b/w two nodes.
It can provide connection authentication, transmission encryption, compression.

PPP is used over many types of physical network including serial cable, phone line, trunk line, cellular telephone, radio links, fibre optic links such as SONET.

Two derivate of PPP are : (i) PPPoE (over Internet)
(ii) PPPoA (over ATM)
are used most commonly by ISPs to establish a Digital Subscriber Line (DSL) Internet service connection.

PPP provides error detection, supports multiple protocols.

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Telnet (Level- Remote Login)
is a networking protocol and software program used to access remote computers and terminals over Internet or a TCP/IP computer network.
Using telnet client establishes a connection to remote host, your client becomes a virtual terminal allowing you to communicate with remote host from your computer.

In short, Through telnet, an administrator or another user can access someone else's computer remotely.
It is a network protocol used on Internet or local area network (LAN) to provide a bidirectional interactive communication facility.

Important port.
Port 21 – File Transfer Protocol
Port 23 – Telnet server
Port 25 – SMTP (simple mail transfer protocol)
Port 53 – DNS (Domain Name server)
Port 80 – HTTP
Port 110 – POP3 (Post office protocol 3)

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GSM (Global system for Mobile communication)
- it is a digital cellular technology used for transmitting mobile voice and data services.
- GSM emerged at Bell Laboratories at 1970s.
- It is a circuit-switched system that divides each 200kHz channel into eight 25 kHz time slots.
- It makes use of TDMA (Time division multiple Access) to transmit signals
- GSM digitizes and compresses data, sends it down through a channel with two other streams of user data, each in its own timeslot.

advantages.
- improved spectrum efficiency
- international roaming
- low cost mobile sets
- compatible with ISDN (Integrated Service Digital Network)

GSM is divided into
(i) Mobile Station (MS) (ii) Base station subsystem (BSS)
(iii) Network switching subsystem (NSS)
(iv) Operation support subsystem (OSS).

IMEI (International Mobile station equipment identity) is a serial no which distinctively identifies a mobile system internationally.

IMSI (International Mobile subscriber Identity)
SIM (subscriber Identity Module)
MSISDN (mobile subscriber ISDN) - authenticate telephone number of mobile station.

CDMA (code division Multiple Access)

- digital cellular technology
- CDMA cellular systems are deemed superior to FDMA and TDMA
- Every channel uses full spectrum available.
- Individual conversations are encoded with pseudo-random digital sequence and then transmitted using a wide frequency range
- allows more subscribers to connect at any given time.
- provides better capacity for voice and data communication.
- uses FDMA and TDMA.

CDMA channels — forward channel and reverse channel.

WLL (wireless local loop)

- is use of wireless communication link as the "last mile/first mile" connection for delivering plain old telephone services (POTS) or Internet Access to telecommunication customers.
- different types of WLL → Broadband wireless Access, Radio in Loop, Fixed Radio Access, Fixed wireless Access.

RF Issues and solution in WLL:

RF services are provided by for licensed frequencies. These frequencies assures that data would be transferred on WLL without any interference. But nowdays some unlicensed frequencies are also used. So to protect rights of licensed frequencies, authentication & encryption are used: with some other solutions are named as LMDS, MMDS, cellular telephony and WLANs.

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LMDS (Local Multipoint Distribution Services) is a licensed microwave frequency (1.5 GHz – 45 GHz bands)

MMDS (Multichannel multipoint distribution service) is a two way communication method. (range 35 miles)

Licensed Microwave is a point to point microwave technology (upto 5 miles)

Cellular telephony has actually achieved status of WLL.

WLANs run in unlicensed frequency bands. WLAN technology includes Wi-Fi, Wi-Fi 5, Bluetooth

Difference b/w CDMA, WLL, GSM, EDGE.

CDMA – used to transmit wireless calls by assigning them codes. Many users share same frequency channel

WLL – is system that connects wireless user to PSTN using wireless technology and other circuitry to complete last mile b/w wireless user and exchange equipment.

GSM – globally accepted standard for digital cellular communication.

EDGE – (Enhanced Data rate for Global evolution) – is a evolutionary step in GSM – development path for faster delivery of data, delivered at rate of 284 kbps.

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3G (Third generation)

In 2005, there were 23 networks worldwide, operating 3G technology. Telenor launched 3G network in Europe.

- it enables increased data-handling rates and high speed bandwidth.
- it is a network protocol which refers to generation of mobile phones and telecommunication equipments which are compatible with International Mobile Telecommunication-2000 (IMT 2000) standards stated by International Telecommunication Union (ITU).
- advanced features like video conferencing calling, video call and enhanced browsing capabilities.
- bandwidth is calculated from difference of high and low frequencies.

Other contributors of 3G:

CDMA 2000 - code division multiple access
TD-SCDMA - Time-division synchronous Code-division Multiple Access.
W-CDMA (UMTS) - wideband code division multiple access.

advantages - use IP connectivity, which is packet based.
- fast download, quality of voice & video calls is high, increased bit rate

disadvantage - costly, high power consumption
- has to pay high amount for 3G licensing
- base station should be closer to access full strength.

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SMS (short message service)
- is a service for sending short messages of upto 160 characters (224 characters if using 5 bit mode)

EMS (Enhanced Messaging service), an adaptation of SMS that allows user to send and receive ringtones and operator logos.

SMS gateways are Websites that allow users to send messages to people within cell served by that gateway.

- SMS service is now available in all wireless network such as GSM, TDMA, CDMA etc.
- SMS has store & forward feature
- SMS is sent to one or more receiver, an acknowledgement is guaranteed.

Electronic Mail (E-mail)
- Consist of message which are sent and received via Internet.
- messages are delivered within minutes or seconds.
- sender and receiver don't have to be working at same time.
- keep record of communication
- reach lot of people at once.

disadvantage:
- junk mail, ads., misinterpretation, messages can be passed onto others

understanding email address.
- one should have email account and email address.
Email address → username @ email provider's domain.

Email provider is website that hosts your email account.

Top three providers → Yahoo, outlook, gmail.

Web domain → .com (mainly used for commercial), .gov (government websites), .edu (schools), .mil (military branches), .org (non-profit organizations).

Chat

- process of communicating, interacting, exchanging messages over Internet.
- involves two or more individuals that communicate through chat enabled service or software.
- may be delivered through text, verbal, audio, visual etc communication via internet.
- if connected through dekstop, chat requires software that supports IRC (Internet Relay Chat) or an instant messaging application, where a central server manages chat communication b/w different end user clients.

Video Conferencing

- conducting a video conference or video teleconference
- means to conduct a conference b/w two or more participants at different location by using computer network to transmit audio and video data.
- Multipoint video conferencing allows three or more participants to communicate.

example: Apple's Facetime, Google's Hangout, Microsoft Skype.

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(Cyber law (Internet law) (Cyberspace) (IT Act 2000)
law that deals with Internet's relationship to technological and electronic elements. It prevent or reduce large scale damage from cyber criminal activities by protecting information access, privacy, communications, intellectual property (IP) and freedom of speech related to user of Internet, websites, email etc.

cyber crimes can be categorised in two ways:

(i) Computer as a target → using computer to attack other computer. eg: Hacking, Virus etc.

(ii) Computer as a weapon → using computer to commit real world crimes. eg: Credit card frauds, Cyber Terrorism, EFT frauds etc.

Firewall (Personal firewalls)
may be defined as single point b/w two or more network where all traffic must pass, the firewall authenticates, controls and logs all traffic.

Firewall is a hardware or software based network system that controls incoming and outgoing network traffic by analysing data packets and determining whether they should be allowed through or not based on certain rule set. It can have a variety of configuration depending upon user requirement. It builds a bridge b/w internal and external network and protects against →

- remote login
- application back doors
- virus, spam, malware etc.

They run on single computer
A software firewall prevents unwanted access to computer by identifying, prevent communication over risky ports.

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Types of Firewall.

Traffic controlling technique Firewall can be categorised as:

(I) **Packet filtering firewall:**

- operates by filtering incoming & outgoing packets using routers or devices that have been configured to screen.

It checks - source and destination address,
application or protocol
source port no and destination port no.

(II) **Circuit level firewall.**

- along with allowing & disallowing packets, it determines whether connection b/w both ends is valid. It validates each connection session establishment. for exchange of data and monitors TCP handshaking.

It checks - source and destination address and port no
- time of day
- protocol
- user and password.

(III) **Application level gateway** (also called proxies)

- similar to circuit level but is application specific. ie. it can filter packets at application layer of OSI model. Incoming & outgoing packets are unable to access services for which there is no proxy. It can also be used to log user activity and logins. They can also filter application specific commands.

(IV) **Stateful Multilayer Inspection:**

combination of all three. Traffic is filtered based on specific application rules, such as specified app,

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browser or protocal (FTP) or combinations. It filters packets at network layer which determines whether a session is legitimate or not and then evaluates the content of packet at application layer.

Cookies

Cookies are small files which are stored on user's computer. They are designed to hold a modest amount of data specific to particular client and website, and can be accessed either by web server or client computer. Cookie is able to carry information from one visit to website to next.

A cookie is a information that a website puts on your harddisk so that it can remember something about you at later time. A cookie is a mechanism that allows server to store its information about a user on user's own computer. Internet Explorer stores each cookie as a seperate file under a windows subdirectory.
Netscape stores in → single cookies.txt file.
Opera stores in → single cookies.dat file.

Authentication cookies are most common method used by web servers to know whether user is logged in or not, which account they are logged in with.

Hackers (Ethical Hackers) (white hat hacker)

is a person who is extremely intrested in explaring the things. Hackers are expert programmers.
Ethical hacking meanings you think like Hackers, first you hack system and find out loop holes and then try to correct those loop holes. These type of hackers protect cyberworld from every possible threat and severe future

from coming security loop holes.

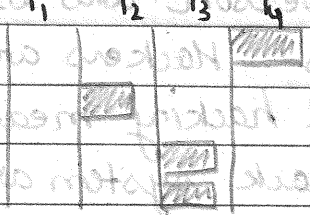
(rackers (Black hat) (cheaters) (Phishers)
they are criminals because they have mindset of causing harm to security to steal useful data and use it in wrong ways.
phishers also steal account info. and steal your credit card no, money over net.

Important points.

(I) Repeater, Hub → passive device, one layer switch.
(II) Bridge → two layer
(III) Router - three layer
(IV) Gateway - all layer device
→ active device.

ALOHA - is a part of data link layer.
protocols

Pure aloha	slotted Aloha.
Out of 100 frames that are transmitted, only max 18.4 frames are transmitted will reach destination succesfully. So, throughput → 18.4 %	Aloha is divided into time slots and each sender will send in its time slot. throughput → 36.8 % ie. max 36.8 out of 100 will be transmitted succesfully.



Throughput → rate at which user transmit data and data should safely reach destination.